Differential Equations

Calculus Lecture Notes

Dr. Nguyen Van Hoi-UIT

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Outline

First-Order Linear Differential Equations

Standard Form:

$$y' + p(x)y = q(x)$$

Steps to Solve:

1. Calculate the Integrating Factor:

$$I(x) = e^{\int p(x)dx}$$

2. Multiply both sides by $I(x)$:

$$(y \cdot I(x))' = q(x) \cdot I(x)$$

3. Integrate both sides and solve for y:

$$y \cdot I(x) = \int q(x)I(x)dx + C$$

Example 1

Problem: Solve $y' + 3x^2y = 6x^2$.

Solution:

1. Integrating Factor:

$$I(x) = e^{\int 3x^2 dx} = e^{x^3}$$

2. Multiply equation by e^{x^3}:

$$y'e^{x^3} + 3x^2e^{x^3}y = 6x^2e^{x^3} \implies (ye^{x^3})' = 6x^2e^{x^3}$$

3. Integrate both sides:

$$ye^{x^3} = \int 6x^2e^{x^3}dx = 2e^{x^3} + C \implies \mathbf{y = 2 + Ce^{-x^3}}.$$

Example 2: Initial Value Problem

Problem: Solve $x^2y' + xy = 1$, $x > 0$, $y(1) = 2$.
Solution: Rewrite in standard form: $y' + \frac{1}{x}y = \frac{1}{x^2}$.

- $I(x) = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$.
- Multiply by x:
$$(yx)' = \frac{1}{x}$$
- Integrate:
$$yx = \int \frac{1}{x}dx = \ln x + C \implies y = \frac{\ln x + C}{x}$$
- Use $y(1) = 2$:
$$2 = \frac{\ln 1 + C}{1} \implies C = 2 \implies y = \frac{\ln x + 2}{x}.$$

Example 3

$tu' - 3u = t^2, \quad u(2) = 4$. **Standard Form:** $u' - \frac{3}{t}u = t$.

Integrating Factor:

$$I(t) = e^{\int -\frac{3}{t}dt} = e^{-3\ln t} = t^{-3} = \frac{1}{t^3}.$$

Multiply and Integrate:

$$\left(u \cdot t^{-3}\right)' = t \cdot t^{-3} = t^{-2}$$

$$ut^{-3} = \int t^{-2}dt = \frac{t^{-1}}{-1} + C = -\frac{1}{t} + C \implies u(t) = -t^2 + Ct^3.$$

Find C: Since $u(2) = 4$:

$$4 = -(2)^2 + C(2)^3 \implies 4 = -4 + 8C \implies 8C = 8 \implies \mathbf{C = 1}$$

Result: $u(t) = t^3 - t^2$.

Example 4

$xy' - y = x^2 \sin x,\ \ y(\pi) = 2\pi$ **Standard:** $y' - \frac{1}{x}y = x \sin x.$

Integrating Factor:

$$I(x) = e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$$

Multiply and Integrate:

$$\left(\frac{y}{x}\right)' = (x \sin x) \cdot \frac{1}{x} = \sin x$$

$$\frac{y}{x} = \int \sin x \, dx = -\cos x + C \implies y = x(-\cos x + C)$$

Find C: Since $y(\pi) = 2\pi$:

$$2\pi = \pi(-\cos \pi + C) = \pi(-(-1) + C) = \pi(1 + C) \implies \mathbf{C = 1}$$

Result: $y = x(1 - \cos x)$.

Exercises

1. $y' + y = x$.
   Hint
   $I(x) = e^x \Rightarrow (ye^x)' = xe^x$.
   Integrate by parts: $ye^x = xe^x - e^x + C$.
   $\mathbf{y = x - 1 + Ce^{-x}}$.
2. $xy' + y = \sqrt{x}$.
   Hint
   Notice $(xy)' = x^{1/2}$.
   $xy = \frac{2}{3}x^{3/2} + C \implies \mathbf{y = \frac{2}{3}\sqrt{x} + \frac{C}{x}}$.
3. $(x^2 + 1)y' + xy = 1,\ y(0) = 2$.
4. $y' + 3y = 2xe^{-3x},\ y(0) = 1$.
5. $y' - 2xy = e^{x^2}$.

Bernoulli Differential Equations

A first-order differential equation is of **Bernoulli type** if it can be written as:

$$y' + P(x)y = Q(x)y^n$$

where $n \in \mathbb{R}$ and $n \neq 0, 1$.

Method of Solution:

1. Divide by y^n: $y^{-n}y' + P(x)y^{1-n} = Q(x)$.
2. Substitute $u = y^{1-n}$. Then $u' = (1-n)y^{-n}y'$.
3. Convert to linear equation for u:

$$\frac{u'}{1-n} + P(x)u = Q(x) \implies u' + (1-n)P(x)u = (1-n)Q(x)$$

4. Solve for u (linear), then substitute back $y = u^{\frac{1}{1-n}}$.

Example 1

Problem: Solve $y' + \frac{1}{x}y = xy^2$.
Solution: Here $n = 2$, $P(x) = 1/x$, $Q(x) = x$.

1. Divide by y^2: $y^{-2}y' + \frac{1}{x}y^{-1} = x$.
2. Let $u = y^{1-2} = y^{-1}$. Then $u' = -y^{-2}y' \implies -u' = y^{-2}y'$.
3. Substitute: $-u' + \frac{1}{x}u = x \implies u' - \frac{1}{x}u = -x$.
4. Solving linear eq:
    - Integrating factor: $I(x) = e^{\int -1/x dx} = e^{-\ln x} = 1/x$.
    - $(u \cdot \frac{1}{x})' = -x \cdot \frac{1}{x} = -1$.
    - Integrate: $u/x = -x + C \implies u = -x^2 + Cx$.
5. Back-substitute $u = 1/y$:

$$\frac{1}{y} = -x^2 + Cx \implies \mathbf{y} = \frac{\mathbf{1}}{\mathbf{Cx - x^2}}$$

Example 2

Problem: Solve $xy' + y = x^2y^3, \quad x > 0$. **Solution:** Standard form: $y' + \frac{1}{x}y = xy^3$. $(n = 3)$.

1. Divide by y^3: $y^{-3}y' + \frac{1}{x}y^{-2} = x$.
2. Let $u = y^{1-3} = y^{-2}$. Then $u' = -2y^{-3}y'$.
3. Substitute: $\frac{u'}{-2} + \frac{1}{x}u = x \implies u' - \frac{2}{x}u = -2x$.
4. Integrating factor: $I(x) = e^{\int -2/x dx} = x^{-2}$.
5. Solve for u:
$$(ux^{-2})' = -2x \cdot x^{-2} = -2/x$$
$$ux^{-2} = -2\ln x + C \implies u = x^2(C - 2\ln x)$$
6. Result: $\mathbf{y^{-2} = x^2(C - 2\ln x)} \implies y = \pm\frac{1}{x\sqrt{C - 2\ln x}}$.

Example 3

Problem: Solve the differential equation:

$$y' + \frac{2}{x+1}y + (x+1)^3 y^2 = 0.$$

Rewrite in Bernoulli Form:

$$y' + \frac{2}{x+1}y = -(x+1)^3 y^2.$$

This is a **Bernoulli equation** of the form $y' + P(x)y = Q(x)y^n$:

- $P(x) = \frac{2}{x+1}$.
- $Q(x) = -(x+1)^3$.
- $n = 2$.

Since $n = 2$, we divide the equation by y^2 (assuming $y \neq 0$):

$$y^{-2}y' + \frac{2}{x+1}y^{-1} = -(x+1)^3.$$

Let $u = y^{1-n} = y^{1-2} = y^{-1}$.
Then, differentiate u with respect to x:

$$u' = \frac{du}{dx} = -y^{-2}y' \implies y^{-2}y' = -u'$$

Substitute into the equation:

$$-u' + \frac{2}{x+1}u = -(x+1)^3 \Leftrightarrow u' - \frac{2}{x+1}u = (x+1)^3.$$

This is now a **Linear Differential Equation**.

For the linear equation $u' + P_1(x)u = Q_1(x)$, where $P_1(x) = -\frac{2}{x+1}$. Calculate the **Integrating Factor (IF)**:

$$\begin{aligned}\mu(x) &= e^{\int P_1(x)dx} = e^{\int -\frac{2}{x+1}dx} \\ &= e^{-2\ln|x+1|} = e^{\ln((x+1)^{-2})} \\ &= \frac{1}{(x+1)^2}.\end{aligned}$$

Multiply both sides of Eq. (13) by $\mu(x)$:

$$\frac{u'}{(x+1)^2} - \frac{2u}{(x+1)^3} = \frac{(x+1)^3}{(x+1)^2}.$$

$$\frac{d}{dx}\left[\frac{u}{(x+1)^2}\right] = x+1.$$

Integrate both sides with respect to x:

$$\begin{aligned}\int \frac{d}{dx}\left[\frac{u}{(x+1)^2}\right]dx &= \int (x+1)dx \\ \frac{u}{(x+1)^2} &= \frac{(x+1)^2}{2} + C.\end{aligned}$$

Solve for u:

$$u = \left[\frac{(x+1)^2}{2} + C\right](x+1)^2$$

$$u = \frac{(x+1)^4}{2} + C(x+1)^2.$$

Substitute back $y = u^{-1} = \frac{1}{u}$.
The general solution is:

$$y = \frac{1}{\frac{(x+1)^4}{2} + C(x+1)^2}.$$

Simplifying the fraction:

$$y = \frac{2}{(x+1)^4 + 2C(x+1)^2}.$$

(Note: $y = 0$ is also a singular solution to the original equation).

Exercises: Bernoulli Equations

Solve the following differential equations:

1. $y' + y = y^2$ *Hint:* $n = 2, u = y^{-1}$. *Ans:* $y = \frac{1}{1+Ce^x}$
2. $y' - \frac{2}{x}y = \frac{y^2}{x^3}$ *Hint:* $n = 2, u = y^{-1}$. *Ans:* $y = \frac{5x^2}{1+Cx^5}$ *?? Check calculation:* $u' + (2/x)u = -1/x^3 \dots$
3. $x^2y' + 2xy = 5y^4$ *Hint: Rewrite* $y' + \frac{2}{x}y = \frac{5}{x^2}y^4$. *Then* $n = 4, u = y^{-3}$.

Separable Equation

A first-order differential equation is called **separable** if it can be written in the form:

$$\frac{dy}{dx} = g(x)h(y).$$

Key idea: We can separate the variables so that all terms involving y are on one side and all terms involving x are on the other.

$$\frac{1}{h(y)}dy = g(x)dx.$$

Method of Solution

1. **Separate the variables:** Move all y terms to the left (with dy) and all x terms to the right (with dx).

$$N(y)dy = M(x)dx.$$

2. **Integrate both sides:**

$$\int N(y)dy = \int M(x)dx + C.$$

3. **Solve for y (if possible):** Manipulate the resulting equation to find the explicit solution $y = y(x)$.

Example Problem

Problem: Find the general solution of the differential equation:

$$\frac{dy}{dx} = -4xy^2.$$

Step 1: Separate the variables
Assume $y \neq 0$, divide by y^2 and multiply by dx:

$$\frac{1}{y^2}dy = -4xdx.$$

Rewrite as powers:

$$y^{-2}dy = -4xdx.$$

Step 2: Integrate both sides

$$\int y^{-2} dy = \int -4x dx.$$

Using the power rule $\int u^n du = \frac{u^{n+1}}{n+1}$:

$$\frac{y^{-1}}{-1} = -4\left(\frac{x^2}{2}\right) + C \implies -\frac{1}{y} = -2x^2 + C.$$

Step 3: Solve for y Multiply by -1:

$$\frac{1}{y} = 2x^2 - C.$$

Let $C_1 = -C$ (still an arbitrary constant):

$$y = \frac{1}{2x^2 + C_1}.$$

Note on Singular Solutions: In Step 1, we divided by y^2, assuming $y \neq 0$. We must check if $y = 0$ is a solution to original equation $y' = -4xy^2$.

- LHS: $y' = 0$.
- RHS: $-4x(0)^2 = 0$.

Since LHS $=$ RHS, $y = 0$ **is also a solution** (singular solution).

Practice Exercises

Solve the following separable differential equations:

1. **Exercise 1:** Find the general solution of:

$$\frac{dy}{dx} = \frac{x^2}{y}.$$

2. **Exercise 2:** Find the general solution of:

$$y' = e^{x-y}.$$

3. **Exercise 3 (Initial Value Problem):**

$$\frac{dy}{dx} = y\cos x, \quad \text{with } y(0) = 1.$$

Homogeneous

Definition: A first-order differential equation is **homogeneous** if it can be written in the form:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right).$$

Method of Solution:

1. **Substitution:** Let $y = vx$, where $v = v(x)$ is a function of x.
2. **Derivative:** Using the product rule:

$$y' = v + x\frac{dv}{dx}.$$

3. **Transform:** Substitute y and y' into the original equation. The new equation in terms of v and x will always be *separable*.

Example 1

Solve: $2xy\frac{dy}{dx} = x^2 + 2y^2$

Step 1: Rewrite in form $f(y/x)$

$$\frac{dy}{dx} = \frac{x^2 + 2y^2}{2xy} = \frac{x}{2y} + \frac{y}{x} = \frac{1}{2}\left(\frac{y}{x}\right)^{-1} + \left(\frac{y}{x}\right).$$

Step 2: Substitution Let $y = vx \implies y' = v + xv'$.

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{1}{2v} + v \\ x\frac{dv}{dx} &= \frac{1}{2v} \quad \text{(Cancel } v \text{ on both sides).} \end{aligned}$$

Step 3: Separate Variables

$$2v\,dv = \frac{dx}{x}.$$

Step 4: Integrate

$$\int 2v\,dv = \int \frac{dx}{x}.$$

$$v^2 = \ln|x| + C.$$

Step 5: Back-substitution Recall that $v = \frac{y}{x}$.

$$\left(\frac{y}{x}\right)^2 = \ln|x| + C.$$

Multiply by x^2 to get the general solution:

$$y^2 = x^2(\ln|x| + C).$$

Example 2

Solve: $xy' = y + 2\sqrt{xy}$ (Assume $x, y > 0$)

Step 1: Rewrite in form $f(y/x)$ Divide by x:

$$y' = \frac{y}{x} + \frac{2\sqrt{xy}}{x} = \frac{y}{x} + 2\sqrt{\frac{y}{x}}.$$

Step 2: Substitution Let $y = vx \implies y' = v + xv'$.

$$\begin{aligned} v + x\frac{dv}{dx} &= v + 2\sqrt{v} \\ x\frac{dv}{dx} &= 2\sqrt{v} \quad \text{(Cancel } v\text{)}. \end{aligned}$$

Step 3: Separate Variables

$$\frac{dv}{2\sqrt{v}} = \frac{dx}{x}.$$

Step 4: Integrate Rewrite LHS as $\frac{1}{2}v^{-1/2}dv$:

$$\int \frac{1}{2\sqrt{v}}\,dv = \int \frac{dx}{x}.$$

$$\sqrt{v} = \ln|x| + C.$$

(Note: derivative of $\sqrt{v}$ is $\frac{1}{2\sqrt{v}}$).
Step 5: Back-substitution Recall $v = \frac{y}{x}$:

$$\sqrt{\frac{y}{x}} = \ln|x| + C.$$

Square both sides:

$$y = x(\ln|x| + C)^2.$$

Example 3

Solve the differential equation:

$$y' = \frac{y}{x} + x^2 e^{-x} \cos^2\left(\frac{y}{x}\right).$$

Step 1: Substitution Let $y = vx \implies v = \frac{y}{x}$.
Differentiate with respect to x:

$$y' = v + xv'.$$

Substitute into the original equation:

$$v + xv' = v + x^2 e^{-x} \cos^2(v).$$

Cancel v from both sides:

$$xv' = x^2 e^{-x} \cos^2(v).$$

Divide by x (assuming $x \neq 0$):

$$v' = xe^{-x} \cos^2(v).$$

Step 3: Separate Variables Rewrite v' as $\frac{dv}{dx}$:

$$\frac{dv}{dx} = xe^{-x} \cos^2(v).$$

Move terms involving v to the left and x to the right:

$$\frac{dv}{\cos^2(v)} = xe^{-x} dx.$$

Integrate both sides of (30):

$$\int \frac{1}{\cos^2(v)} dv = \int xe^{-x} dx.$$

Left Hand Side (LHS): Recall that $\frac{1}{\cos^2 v} = \sec^2 v$.

$$\text{LHS} = \int \sec^2 v \, dv = \tan v.$$

Right Hand Side (RHS): We need to calculate $\int xe^{-x} dx$. This requires **Integration by Parts**.

Calculate: $I = \int xe^{-x}dx$
Use formula $\int u\,dw = uw - \int w\,du$.

- Let $u = x \implies du = dx$.
- Let $dw = e^{-x}dx \implies w = -e^{-x}$.

Apply the formula:

$$\begin{aligned} I &= x(-e^{-x}) - \int (-e^{-x})dx \\ &= -xe^{-x} + \int e^{-x}dx \\ &= -xe^{-x} - e^{-x} + C \\ &= -e^{-x}(x+1) + C. \end{aligned}$$

Combine LHS and RHS:

$$\tan v = -e^{-x}(x+1) + C.$$

Back-substitution: Replace v with $\frac{y}{x}$:
Final Answer

$$\tan\left(\frac{y}{x}\right) = -e^{-x}(x+1) + C.$$

(Or written as: $\frac{y}{x} = \arctan\left[C - e^{-x}(x+1)\right]$)

Exact Differential Equation

A differential equation of the form:

$$P(x, y)dx + Q(x, y)dy = 0.$$

is called an **Exact Differential Equation** if there exists a function $F(x, y)$ such that:

$$\frac{\partial F}{\partial x} = P(x, y) \quad \text{and} \quad \frac{\partial F}{\partial y} = Q(x, y).$$

Test for Exactness: Equation is exact if and only if:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Method of Solution

1. **Check Exactness:** Verify that $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.
2. **Integrate P with respect to x:**
$$F(x, y) = \int P(x, y)dx + g(y).$$
(Here $g(y)$ is the constant of integration with respect to x).
3. **Differentiate with respect to y:**
$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}\left(\int Pdx\right) + g'(y).$$
4. **Solve for g(y):** Set $\frac{\partial F}{\partial y} = Q(x, y)$ to find $g'(y)$, then integrate to find $g(y)$.
5. **Final Answer:** Write $F(x, y) = C$.

Example 1

Solve: $(4x - y)dx + (6y - x)dy = 0.$

Step 1: Check Exactness. Here $P = 4x - y$ and $Q = 6y - x$.

$$\frac{\partial P}{\partial y} = -1, \quad \frac{\partial Q}{\partial x} = -1 \implies \text{Exact!}$$

Step 2: Find F(x,y) Integrate P with respect to x:

$$F(x, y) = \int (4x - y)dx = 2x^2 - xy + g(y).$$

Step 3: Find g(y) Differentiate F w.r.t y and equate to Q:

$$\frac{\partial F}{\partial y} = -x + g'(y) = 6y - x \implies g'(y) = 6y.$$

$$\implies g(y) = \int 6y dy = 3y^2.$$

Substitute $g(y) = 3y^2$ back into $F(x, y)$.
The potential function is:

$$F(x, y) = 2x^2 - xy + 3y^2.$$

General Solution

$$2x^2 - xy + 3y^2 = C.$$

Example 2

Solve: $(2xy^2 + 3x^2)dx + (2x^2y + 4y^3)dy = 0.$

Step 1: Check Exactness. Here $P = 2xy^2 + 3x^2$ and $Q = 2x^2y + 4y^3$.

$$\frac{\partial P}{\partial y} = 4xy, \quad \frac{\partial Q}{\partial x} = 4xy \implies \text{Exact!}$$

Step 2: Integrate P w.r.t x

$$F = \int (2xy^2 + 3x^2)dx = x^2y^2 + x^3 + g(y).$$

Step 3: Differentiate w.r.t y

$$\frac{\partial F}{\partial y} = 2x^2y + g'(y).$$

Step 4: Equate to Q

$$2x^2y + g'(y) = 2x^2y + 4y^3$$

$$g'(y) = 4y^3$$

$$g(y) = \int 4y^3 dy = y^4.$$

Step 5: Final Solution Substitute $g(y)$ back into F:
General Solution

$$x^2y^2 + x^3 + y^4 = C.$$

Example 3

Problem: Solve the differential equation

$$\left(x^3 + \frac{y}{x}\right) dx + (y^2 + \ln x)dy = 0.$$

Step 1: Check Exactness. Here $P(x, y) = x^3 + \frac{y}{x}$ and $Q(x, y) = y^2 + \ln x$.
Compute partial derivatives:

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}\left(x^3 + \frac{y}{x}\right) = \frac{1}{x}$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y^2 + \ln x) = \frac{1}{x}.$$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, the equation is **Exact**.

Step 2: Find Potential Function $F(x, y)$ Integrate P with respect to x:

$$F(x,y) = \int P(x,y)dx = \int \left(x^3 + \frac{y}{x}\right) dx$$
$$= \frac{x^4}{4} + y\ln x + g(y).$$

Step 3: Determine $g(y)$ Differentiate F with respect to y and equate to Q:

$$\frac{\partial F}{\partial y} = \ln x + g'(y) = y^2 + \ln x \implies g'(y) = y^2$$

$$\implies g(y) = \int y^2 dy = \frac{y^3}{3}.$$

Substitute $g(y)$ back into $F(x, y)$. The general solution is $F(x, y) = C$:

$$\frac{x^4}{4} + y \ln x + \frac{y^3}{3} = C.$$

Integrating Factors (Theory)

Problem: What if $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$?

Concept: We look for a function $\mu(x, y)$ (called an *Integrating Factor*) such that multiplying the equation by μ makes it exact:

$$\mu(x, y)Pdx + \mu(x, y)Qdy = 0 \quad \text{(Exact)}$$

Finding μ (Two common cases):

1. If $\frac{P_y - Q_x}{Q}$ depends only on x, then $\mu(x) = e^{\int \frac{P_y - Q_x}{Q} dx}$.
2. If $\frac{Q_x - P_y}{P}$ depends only on y, then $\mu(y) = e^{\int \frac{Q_x - P_y}{P} dy}$.

Example: Finding Integrating Factor

Problem: $2xydx + (y^2 - 3x^2)dy = 0$

Step 1: Check Exactness

$$P = 2xy \implies P_y = 2x.$$

$$Q = y^2 - 3x^2 \implies Q_x = -6x.$$

Since $2x \neq -6x$, it is **NOT Exact**.

Step 2: Check for Integrating Factor Try the expression involving P (function of y):

$$\frac{Q_x - P_y}{P} = \frac{-6x - 2x}{2xy} = \frac{-8x}{2xy} = \frac{-4}{y}.$$

This depends only on y. So, an integrating factor exists!

Step 3: Calculate $\mu(y)$

$$\mu(y) = e^{\int \frac{Q_x - P_y}{P} dy} = e^{\int -\frac{4}{y} dy}.$$

$$\mu(y) = e^{-4 \ln|y|} = e^{\ln(y^{-4})} = y^{-4} = \frac{1}{y^4}.$$

Multiplying the original equation by $\frac{1}{y^4}$ will convert it into an exact differential equation, which can then be solved using the standard method.

Example 1

Equation:

$$(xy + y^2)dy + y^2 dx = 0.$$

Rearrange to isolate $\frac{dx}{dy}$:

$$y^2 dx = -(xy + y^2)dy.$$

$$\frac{dx}{dy} = -\frac{xy + y^2}{y^2} = -\left(\frac{x}{y} + 1\right).$$

This is a **Homogeneous Equation** with respect to y. We use the substitution from the notes (Method C_2).

Step 1: Substitution Let $x = uy$ (where u is a function of y). Differentiate with respect to y:

$$\frac{dx}{dy} = u'y + u.$$

Substitute into the differential equation:

$$u'y + u = -(u+1)$$

$$u'y = -u - 1 - u$$

$$u'y = -2u - 1.$$

Step 2: Separation of Variables

$$y\frac{du}{dy} = -(2u+1)$$

$$\frac{du}{2u+1} = -\frac{dy}{y}.$$

Step 3: Integrate both sides

$$\int \frac{du}{2u+1} = -\int \frac{dy}{y}$$

$$\frac{1}{2}\ln|2u+1| = -\ln|y| + C.$$

Multiply by 2:

$$\ln|2u+1| = -2\ln|y| + 2C$$

$$\ln|2u+1| + \ln(y^2) = K.$$

Step 4: Back-substitution Combine logs: $\ln |(2u+1)y^2| = K$.
Exponentiate: $(2u+1)y^2 = C_1$.
Substitute $u = \frac{x}{y}$:

$$\left(2\frac{x}{y}+1\right)y^2 = C_1$$

$$2xy + y^2 = C_1.$$

Final Answer

$$y^2 + 2xy = C.$$

Example 2

Equation:

$$(2y^2 - 2xy + x^2)dx - xydy = 0.$$

Rewrite as:

$$\frac{dy}{dx} = \frac{2y^2 - 2xy + x^2}{xy}.$$

Divide numerator terms by xy:

$$\frac{dy}{dx} = 2\frac{y}{x} - 2 + \frac{x}{y}.$$

This is a **Homogeneous Equation** $(y' = f(y/x))$.

Step 1: Substitution Let $y = ux \implies y' = u'x + u$.
Substitute into the equation:

$$u'x + u = 2u - 2 + \frac{1}{u}.$$

Simplify:

$$u'x = u - 2 + \frac{1}{u}$$

$$u'x = \frac{u^2 - 2u + 1}{u} = \frac{(u-1)^2}{u}.$$

Step 2: Separation of Variables

$$\frac{u}{(u-1)^2}du = \frac{dx}{x}$$

Step 3: Integrate LHS Let $w = u - 1 \implies u = w + 1$.

$$\int \frac{w+1}{w^2}dw = \int \left(\frac{1}{w} + w^{-2}\right) dw$$

$$= \ln|w| - \frac{1}{w} = \ln|u-1| - \frac{1}{u-1}.$$

Equating to RHS ($\int \frac{dx}{x} = \ln|x| + C$):

$$\ln|u-1| - \frac{1}{u-1} = \ln|x| + C.$$

Step 4: Back-substitution Substitute $u = \frac{y}{x}$:

$$\ln\left|\frac{y}{x} - 1\right| - \frac{1}{\frac{y}{x} - 1} = \ln|x| + C.$$

Simplify terms:

$$\ln\left|\frac{y - x}{x}\right| - \frac{x}{y - x} = \ln|x| + C$$

$$\ln|y - x| - \ln|x| - \frac{x}{y - x} = \ln|x| + C.$$

General Solution

$$\ln|y - x| - \frac{x}{y - x} - 2\ln|x| = C.$$

Comprehensive Practice Problems

1. **Bernoulli Equation:**

$$y' + \frac{2}{x+1}y + (x+1)^3 y^2 = 0.$$

2. **Homogeneous Equation:**

$$(x+y)\frac{dy}{dx} = x - y.$$

3. **Exact Equation:**

$$(2xy^2 + 3x^2)dx + (2x^2y + 4y^3)dy = 0.$$

4. **Advanced Homogeneous (Substitution):**

$$y' = \frac{y}{x} + x^2 e^{-x} \cos^2\left(\frac{y}{x}\right).$$

Second-Order Linear Homogeneous Equations

Form: $ay'' + by' + cy = 0$ (a, b, c constants).
Consider $ar^2 + br + c = 0$. Let $\Delta = b^2 - 4ac$.

- **Case 1 ($\Delta > 0$):** Two distinct real roots $r_1 \neq r_2$.

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}.$$

- **Case 2 ($\Delta = 0$):** One repeated root $r_1 = r_2 = r$.

$$y = c_1 e^{rx} + c_2 x e^{rx}.$$

- **Case 3 ($\Delta < 0$):** Complex roots $\alpha \pm i\beta$.

$$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x).$$

Homogeneous Examples

1. Distinct Real Roots: $y'' + y' - 6y = 0$.
$r^2 + r - 6 = 0 \implies r_1 = 2, r_2 = -3$.
Result: $\mathbf{y = c_1 e^{2x} + c_2 e^{-3x}}$.

2. Repeated Root: $4y'' + 12y' + 9y = 0$.
$4r^2 + 12r + 9 = (2r+3)^2 = 0 \implies r = -3/2$.
Result: $\mathbf{y = c_1 e^{-1.5x} + c_2 x e^{-1.5x}}$.

3. Complex Roots: $y'' - 6y' + 13y = 0$.
$r^2 - 6r + 13 = 0 \implies r = 3 \pm 2i \ (\alpha = 3, \beta = 2)$.
Result: $\mathbf{y = e^{3x}(c_1 \cos 2x + c_2 \sin 2x)}$.

Second-Order Non-Homogeneous Equations

Form: $ay'' + by' + cy = G(x)$.
General Solution: $y = y_c + y_p$

- y_c: Complementary solution (solve homogeneous part).
- y_p: Particular solution (guess based on $G(x)$).

Form of $G(x)$	**Form of** y_p
Polynomial $P_n(x)$	$A_n x^n + \cdots + A_0$
Ce^{kx}	Ae^{kx}
$C \sin kx$ or $C \cos kx$	$A \cos kx + B \sin kx$

**Note:* Multiply by x or x^2 if guess duplicates y_c.*

Example: Polynomial RHS

Problem: $y'' + y' - 2y = x^2$.
Step 1 (y_c): $r^2 + r - 2 = 0 \implies r_1 = 1, r_2 = -2$.

$$y_c = c_1 e^x + c_2 e^{-2x}$$

Step 2 (y_p): Guess $y_p = Ax^2 + Bx + C$.
$y_p' = 2Ax + B, \quad y_p'' = 2A$.
Substitute into ODE:

$$2A + (2Ax + B) - 2(Ax^2 + Bx + C) = x^2$$

$$-2Ax^2 + (2A - 2B)x + (2A + B - 2C) = x^2$$

Step 3: Equate coefficients:

- $x^2 : -2A = 1 \implies A = -1/2$.
- $x^1 : 2A - 2B = 0 \implies B = -1/2$.
- $x^0 : 2A + B - 2C = 0 \implies C = -3/4$.

Result: $y = c_1 e^x + c_2 e^{-2x} - \frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}$.

Example: Exponential RHS

Problem: $y'' + y' - 2y = e^{2x}$.
Step 1: $y_c = c_1 e^x + c_2 e^{-2x}$.
Step 2: Guess $y_p = Ae^{2x}$ (since $2 \neq 1, -2$).
$y_p' = 2Ae^{2x}, \quad y_p'' = 4Ae^{2x}$.
Substitute:

$$(4A + 2A - 2A)e^{2x} = e^{2x}$$

$$4Ae^{2x} = e^{2x} \implies A = 1/4$$

Result: $\mathbf{y = c_1 e^x + c_2 e^{-2x} + \frac{1}{4} e^{2x}}$.

Example: Resonance Case

Problem: $y'' - 2y' - 3y = e^{3x}$.
Step 1: $r^2 - 2r - 3 = 0 \implies r = 3, -1$.

$$y_c = c_1 e^{3x} + c_2 e^{-x}$$

Step 2: Normal guess Ae^{3x} fails (duplicates y_c).
New Guess: $y_p = Axe^{3x}$.
$y_p' = Ae^{3x}(1 + 3x)$.
$y_p'' = Ae^{3x}(3 + 3 + 9x) = Ae^{3x}(6 + 9x)$.
Substitute:

$$Ae^{3x}[(6 + 9x) - 2(1 + 3x) - 3x] = e^{3x}$$

$$A[6 - 2] = 1 \implies 4A = 1 \implies A = 1/4$$

Result: $\mathbf{y = c_1 e^{3x} + c_2 e^{-x} + \frac{1}{4} x e^{3x}}$.

Exercises: Homogeneous Equations

Find the general solution of the following equations:

1. $y'' - 4y' + 3y = 0$ *Hint: Roots are* $r = 1, 3$. *Ans:* $y = c_1 e^x + c_2 e^{3x}$
2. $9y'' + 6y' + y = 0$ *Hint: Root* $r = -1/3$. *Ans:* $y = c_1 e^{-x/3} + c_2 x e^{-x/3}$
3. $y'' + 4y' + 5y = 0$ *Hint: Roots* $-2 \pm i$. *Ans:*

   $y = e^{-2x}(c_1 \cos x + c_2 \sin x)$

Exercises: Non-Homogeneous Equations

Find the general solution using Undetermined Coefficients:

1. $y'' - y' - 2y = 4x^2$ *Ans:* $y = c_1 e^{2x} + c_2 e^{-x} - 2x^2 + 2x - 3$
2. $y'' + y = \sin 2x$ *Ans:* $y = c_1 \cos x + c_2 \sin x - \frac{1}{3} \sin 2x$
3. $y'' - 4y' + 4y = e^{2x}$ *Hint:* y_c *has* e^{2x} *and* xe^{2x}. *Use* $y_p = Ax^2 e^{2x}$.

Exercises: Initial Value Problems

Solve the initial value problems:

$$y'' - 6y' + 8y = 0, \quad y(0) = 2, \quad y'(0) = 2$$

Solution Sketch:

- Characteristic eq: $r^2 - 6r + 8 = (r-2)(r-4) = 0$.
- General solution: $y = c_1 e^{2x} + c_2 e^{4x}$.
- Apply conditions:

$$\begin{cases} c_1 + c_2 = 2 \\ 2c_1 + 4c_2 = 2 \end{cases} \implies \begin{cases} c_1 = 3 \\ c_2 = -1 \end{cases}$$

- Result: $\mathbf{y = 3e^{2x} - e^{4x}}$.

Challenge Problem (Superposition)

Find the general solution of:

$$y'' - 2y' - 3y = e^x + x$$

Hint: Use the Principle of Superposition.

1. Solve homogeneous:
   $y'' - 2y' - 3y = 0 \implies y_c = c_1 e^{3x} + c_2 e^{-x}$.
2. Find y_{p1} for RHS e^x: Guess Ae^x.
3. Find y_{p2} for RHS x: Guess $Bx + C$.
4. Combine: $y = y_c + y_{p1} + y_{p2}$.